



# **An Innovative Approach to Face Recognition Attendance System: Sunglasses with YOLO V9 Technology**

Dissertation submitted for the completion of

**M.Sc -Data Science and Analytics**

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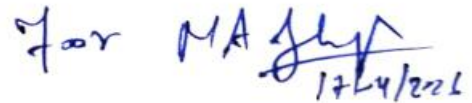
**April 2026**

## **CERTIFICATE**

This is to certify that this dissertation titled “**An Innovative Approach to Face Recognition Attendance System: Sunglasses with YOLO V9 Technology**” Submitted by **Kaushal Sahu** is partial fulfillment of requirements of the degree of Master in Science in Data Science and Analytics of JAIN (Deemed-to-be-University), is based on the result of the research work carried out under the guidance and supervision of Dr. J. Venkata Ramana Raju. This dissertation or any part of this has not been submitted for any purpose to any other university.



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## **DECLARATION**

I affirm that the project work titled “**An Innovative Approach to Face Recognition Attendance System: Sunglasses with YOLO V9 Technology**”, being submitted in partial fulfillment for the award of MASTER OF SCIENCE IN DATA SCIENCE AND ANALYTICS is the original work carried out by me. It has not formed part of any other project work submitted for the award of any degree or diploma, either in this or any other University.



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## **ABSTRACT**

**Introduction:** The rapid evolution of artificial intelligence (AI), particularly in computer vision and deep learning, has fundamentally transformed the creation of autonomous intelligent systems. Many of them incorporate face recognition technology, which is widely used today in applications related to safety, authentication, surveillance, etc. With an ever-increasing desire for automation in routine processes, the integration of various types of AI technology into real-time systems became quite popular. The following research focuses on an innovative way of implementing facial recognition technology by embedding the process into a wearable device to manage attendance. **Background** Current attendance management systems are inefficient because they usually require some time and manual labor, are prone to cheating by proxy attendance, and do not allow much flexibility in terms of implementation. Traditional examples of attendance systems include manual roll call, biometric fingerprint scanners, and RFID-based approaches. Face recognition-based attendance systems also suffer from certain drawbacks since they are limited in terms of portability – they use surveillance cameras installed in specific places, where people should come to be recognized and their attendance registered. Therefore, the development of a convenient, fast, accurate, and efficient attendance system becomes essential in today's world.

**Objectives of the research** To develop a wearable face recognition attendance system using the YOLOv9 architecture. **Implementation** The proposed approach uses a set of smart sunglasses as a wearable device capable of recognizing faces in real-time. The system includes two components: smart glasses and an edge computing platform. The former is equipped with a small HD camera

and allows users to walk freely. Visual information is sent to the latter, where a face recognition pipeline processes data and makes decisions about attendance. An edge computing platform is based on a Raspberry Pi 3 Model B+ and has limited computational power and memory capacity. Despite the mentioned disadvantages, edge computing provides faster decision-making, increased privacy, and reduced reliance on the Internet.

**Creating a facial dataset** In the following section, the process of creating a dataset is described. First, all participants took pictures from various angles, in different light environments, and with different facial expressions. The goal was to create a personalized dataset for the model that would reflect all possible situations that might occur in the real world. All images were preprocessed according to standard machine learning techniques and included appropriate labeling. As a result, the final dataset contained 133170 training images.

**Implementation details** The face recognition pipeline consists of several steps. After obtaining the image frame from the camera, the YOLOv9 model detects faces on them and produces bounding boxes. Next, these regions are fed into the feature extractor that converts facial images into numeric vectors – face embeddings. These embeddings are compared to other vectors stored in the local database to identify people and record their attendance with a timestamp.

**Handling occlusions** One of the most common challenges in face recognition systems is working with partially visible faces due to occlusions like wearing glasses or covering a part of a face with a mask. This issue was addressed during the experiment in order to assess system performance under the mentioned circumstances.

**Experimental results** The results show that the system achieves 90% accuracy while processing images with a speed of 48-50 frames per second (FPS). Thus, the YOLOv9 architecture can be considered as quite effective for implementing face recognition on an embedded device.

**Discussion** In addition to technical details of implementing this approach, certain aspects relating to ethics and privacy need to be discussed. Since there is no centralized data storage, the risk of data breach and leaks decreases to almost zero. Moreover, all operations are performed locally, so there is no need to worry about unauthorized access. Finally, people will have to agree to participate in the experiments beforehand, which increases transparency and responsibility for using the technology.

**Conclusions and Future Work** This dissertation demonstrated a functional wearable attendance system utilizing YOLOv9. The findings confirm that this method is practical for real-world deployment due to its speed, accuracy, and mobility. Moreover, it showed that the YOLOv9 model can successfully perform its tasks despite partial occlusions. Further research could aim at achieving higher accuracy, using cloud-based analytics, or developing applications of the same type in the security and entertainment industries.

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## **CHAPTER 1**

### **INTRODUCTION**

## **INTRODUCTION**

The rise of facial recognition technology stands as a landmark achievement in the fields of artificial intelligence and computer vision. Over the past ten years, the maturation of deep learning—specifically convolutional neural networks (CNNs)—has enabled machines to verify human identities with remarkable precision. As a result, there has been an intensive deployment of face recognition applications for use in security surveillance, access control systems, mobile devices, and smart environments.

At the same time, due to the rising demand for automation in administrative routines, there was a necessity to introduce intelligent systems for managing complex processes. One of the examples is the attendance management, which is crucial for universities, schools, and offices. In this regard, there exist several traditional approaches, such as manual roll calls, signing attendance sheets, and use of card-based systems. Nevertheless, these strategies have a number of flaws, namely inefficiency, susceptibility to manipulation, and low accuracy rates.

Automated systems using facial recognition solve many of these problems but often rely on static camera installations. Such setups are rigid and difficult to scale across varied environments. Wearable technology provides a compelling alternative to these traditional constraints. In particular, a device such as smart glasses or sunglasses equipped with a camera can help create a portable facial recognition system.

However, developing such a system involves navigating hardware constraints, such as the limited resources of a Raspberry Pi, and the high computational demands of real-time processing. The system must also remain effective across changing light levels, different facial orientations, and the presence of accessories like sunglasses.

Apart from technical issues, such projects often involve problems related to ethics, data privacy, safety, and secure storage of biometric data. This paper presents an innovative and highly practical approach to creating a wearable attendance management system with the help of facial recognition technology. Specifically, a hardware prototype will be built and tested with an aim to prove its

usefulness in the context of real-world application. This dissertation is an extended and enhanced version of previously developed research work, incorporating additional implementation details, experimental analysis, and system-level improvements.

## **Design and Implementation**

A wearable facial recognition system is an interesting project from multiple perspectives because of numerous technical challenges and opportunities to integrate various technologies. In particular, one of the main requirements is the need to develop a software part of the solution and select the optimal model to perform the necessary task. Thus, the choice of object detection algorithm is critical for the success of the project.

For this purpose, the YOLOv9-S model was selected because of its ability to strike a balance between computational speed and accuracy. It was designed specifically to run effectively even with resource constraints, while being capable of achieving impressive results in practice. This model was selected after analyzing several other alternatives in order to find the best fit for the needs of the project.

As for the hardware components, Raspberry Pi with an attached micro HD camera was selected as the main building blocks of the system. Such selection was driven by a need to combine high performance and limited energy consumption to create a portable product with minimal environmental impact. In addition to the computing unit and camera, a battery-powered unit will be included as well as additional storage space for local management of data.

The system itself will follow a certain pipeline involving several steps. The first of them will consist of the creation of personal datasets containing facial features of students/teachers/employees. This dataset will include a sufficient number of images taken from different angles to facilitate the recognition process. Then, live video feed will be analyzed to detect faces and record them together with timestamps. Finally, a CSV file will be created with data on attendance.

As for challenges, one of the most interesting aspects of this project is the issue of partial occlusions,

which is common for traditional facial recognition solutions. Nevertheless, the ability to perform adequately even in such conditions allows to distinguish the presented system from similar prototypes.

## **CHAPTER 2**

### **LITERATURE REVIEW**

## **LITERATURE REVIEW**

The problem of development of face recognition-based attendance systems attracts much attention in modern reality due to growing needs of automation and optimization of various activities. Numerous approaches related to deep learning algorithms and computer vision help create reliable and effective systems suitable for various purposes.

### **3.1 Evolution of Face Recognition Systems**

Traditional systems designed for face recognition utilized image processing algorithms that provided an opportunity to identify people and distinguish one another with reasonable accuracy. However, the problem with traditional approaches consisted in inability to overcome variations associated with lighting, pose changes, facial expression modifications, etc. Thus, traditional approaches are ineffective in the context of implementation in real-life situations.

Modern developments related to deep learning and convolutional neural networks allowed overcoming the described limitations. Deep learning approach provided an opportunity for automatic identification of patterns and learning new ones that made face recognition technology highly applicable to various fields including attendance and identity checking [1].

### **3.2 Deep Learning Models for Face Recognition**

Various types of deep learning models designed for face recognition attracted much attention in recent years. Thus, Sinha et al. developed FaceRecNet which represented a highly accurate and lightweight architecture suitable for embedded platforms [7].

Apart from this, the role of pre-trained deep learning models such as ResNet architecture was discussed in numerous articles dedicated to this topic. Such pre-trained models allow extraction of discriminative facial features resistant to changes related to lighting or pose variations. Moreover, application of transfer learning enables better performance of face recognition models trained by means of knowledge from huge datasets.

Thus, development of appropriate deep learning models is a vital component for success of future projects aimed at implementing facial recognition technology.

### **3.3 Real-Time Face Detection and Recognition Systems**

High speed of processing is critical for creation of highly effective and reliable attendance systems as numerous individuals should be detected and recognized instantly without any delays. However, traditional detection approaches were far from being efficient as they could not process information fast enough to provide necessary results within a relatively short period.

Recently, numerous works have been devoted to discussion of single-stage detection systems which allow processing images faster. For example, the latest version of YOLO system called YOLOv9 demonstrated better generalization and performance of the algorithm [5].

Development of fast and efficient systems is crucial for creation of highly reliable face detection and recognition algorithms.

### **3.4 Occlusion and Environmental Resistance**

One of the main obstacles standing in the way of efficient face recognition in real-life conditions is occlusion when some parts of face are covered with various accessories. In this case, face recognition can become difficult as not all features can be clearly distinguished.

Various approaches have been proposed to address the described challenge. Duan et al. reported about development of deep learning models which demonstrate good results in case of occlusion through application of special data augmentation techniques [10].

Another approach consists in implementation of multi-stage detection methods which identify the most important landmarks of face regardless of partial occlusion. Such techniques can be applied successfully in real-life situations.



### **3.5 Multi-Face Detection and Group Recognition**

In many cases, it becomes necessary to identify not just one face within a frame but a number of faces at the same time. Therefore, multi-face recognition becomes a significant issue which should be considered. Currently, it is known that modern detection models are capable of identifying multiple faces quickly and accurately. These models have been developed to cope with crowded frames with varying locations of faces and to detect them successfully [5].

Ability to perform multi-face recognition reliably can be considered as one of the main requirements for effective face recognition systems.

### **3.6 Limitations of Current Methods**

In spite of the described advantages of currently available systems, they have various disadvantages which should be taken into consideration. For example, a vast majority of face detection and recognition systems requires installation of cameras in certain places and utilization of special software. This means that systems become dependent on the infrastructure and lose their ability to detect faces automatically without user interaction. Moreover, current methods face difficulties with detecting distant faces, motion-blurred faces, faces in bad lighting conditions. Apart from this, spoofing can serve as an obstacle to correct functioning of face recognition systems.

The last limitation consists in absence of studies concerning wearable face recognition systems. Most current solutions are aimed at creation of stationary systems while the problem of integration into mobile systems remains unresolved.

### **3.7 Research Gaps and Motivation**

From the analysis of existing literature, it can be observed that although facial recognition-based attendance systems have advanced significantly, several important gaps still exist. These gaps are mainly related to portability, usability, and practical deployment in real-world environments. Most current systems depend on fixed camera installations and centralized setups, which limit their

flexibility. As a result, users are often required to stand in specific locations, making such systems less suitable for dynamic or large-scale environments.

In addition, while many studies focus on improving accuracy and detection performance, relatively few address the development of portable or wearable solutions. Existing systems often rely on high computational power and fixed infrastructure, which restricts their use in everyday scenarios. Moreover, important factors such as handling occlusion, ensuring user convenience, and maintaining data privacy are not always addressed comprehensively.

Another key limitation is the lack of integration between advanced detection models and compact embedded hardware. Although modern models like YOLO offer high speed and accuracy, their implementation in wearable or portable systems is still limited. There is a clear need for research that combines real-time detection capabilities with efficient, lightweight hardware solutions.

These gaps highlight the need for a more flexible attendance system that does not rely on fixed infrastructure and can operate effectively in real-world conditions. Instead of requiring users to adapt to the system, the system should adapt to the user's environment. This requires the development of a lightweight, portable, and user-friendly solution that functions naturally and unobtrusively.

The motivation for this research is to address these challenges by introducing a wearable facial recognition system embedded in sunglasses equipped with a micro HD camera. By utilizing a YOLO-based detection model along with an efficient recognition pipeline, the system aims to deliver real-time performance with high accuracy. Additionally, the use of embedded hardware allows the system to function independently without constant internet connectivity, thereby improving portability and ensuring better data privacy.

## **CHAPTER 3**

### **METHODOLOGY**

## **METHODOLOGY**

This section details the architectural design and execution of the proposed attendance tracking solution. The methodology is broken down into interconnected modules that contribute to the overall effectiveness of the system. It is achieved by dividing the methodology into distinct yet interrelated components that represent the system's major aspects. Each aspect is analyzed separately, which makes it possible to understand the significance and contribution of each component in terms of ensuring the efficiency of the system in question.

The first component that is worthy of analysis is the hardware design. In this regard, it should be noted that the hardware design represents a comprehensive description of the hardware components of the wearable device. It is important to note that the selection process of the hardware components is performed taking into account their main attributes (e.g., size, power consumption). In addition, it is noteworthy that the connection between the hardware components is described thoroughly as it plays an integral part in forming the final product.

Another important aspect that is worth analyzing is the process of preparing the dataset. As a rule, it includes collecting facial images, processing them, and augmenting them. It is important to note that this step is absolutely critical for creating an accurate facial recognition model. In addition, it is crucial to note that the augmentation process contributes significantly to improving the quality of the resulting images, thus increasing the efficiency of facial recognition models. After preparing the dataset, one should move on to discussing the model architecture. In this regard, it should be noted that this component represents a detailed description of the structure of the deep learning model. More specifically, the deep learning model comprises several layers (fully-connected layers, convolutional layers, etc.). Moreover, it is important to mention that the model architecture implies selecting a particular neural network type, which is also discussed in this component.

The last but not least important component is the face recognition pipeline. In this case, it is necessary to pay attention to describing the detection process. To detect faces, one needs to capture their images, pre-process them, input them into the model, and receive the corresponding output. Finally, the face recognition pipeline is designed to perform face verification and recognition

operations in real time.

Thus, all the components mentioned above play an undoubtedly vital role in maintaining the system's overall efficiency. First, it is important to mention that the accuracy of facial recognition can be guaranteed only by optimizing each component's performance. Second, it is important to emphasize that each component is responsible for providing the system with unique functions that are essential for its efficient functioning. Third, the integration of each component into a system guarantees the achievement of real-time performance.

In addition, it is important to note that the thorough description of each component of the system facilitates gaining a deep understanding of its capabilities and limitations. Consequently, the process of making future modifications to the proposed system will be much easier. Thus, it is safe to say that the described methodology made it possible to design a highly efficient attendance system based on facial recognition technology.

#### **4.1 System Overview**

The proposed system is designed as a wearable, real-time facial recognition-based attendance management solution that integrates both hardware and software components into a unified framework. Unlike conventional attendance systems that rely on fixed cameras or manual input methods, this system introduces a portable and user-friendly approach by embedding the entire functionality within a pair of smart sunglasses. The primary objective of this design is to ensure seamless, efficient, and automated attendance tracking without interrupting the natural flow of the environment. At its core, the system operates by capturing live video input through a micro HD camera mounted on the sunglasses. This camera continuously records the surrounding environment from the user's perspective, providing a real-time stream of visual data. The placement of the camera is carefully chosen to ensure that faces are captured from a frontal or near-frontal angle, which significantly improves the accuracy of detection and recognition processes.

Once the video stream is captured, it is processed by an embedded computing unit, specifically a Raspberry Pi, which serves as the central processing component of the system. The Raspberry Pi is

responsible for executing all computational tasks, including running the deep learning model for face detection and managing the subsequent recognition pipeline. The choice of this hardware platform is based on its balance between computational capability, cost-effectiveness, and energy efficiency, making it suitable for wearable applications.

The system follows a structured workflow that enables efficient real-time operation. Initially, each frame from the video stream is passed to the YOLOv9-based face detection module. This module identifies and localizes human faces within the frame by generating bounding boxes around detected regions. The use of YOLOv9 allows the system to perform detection in a single pass, ensuring minimal delay and high processing speed. After detecting faces, the system isolates each detected region and forwards it to the feature extraction stage. In this stage, deep learning-based models convert facial images into numerical feature vectors, commonly referred to as embeddings. These embeddings capture the unique characteristics of an individual's face in a compact mathematical representation, enabling efficient comparison and recognition.

The next stage involves matching the extracted embeddings with a pre-existing database of stored facial features. This database is created during the dataset preparation phase and contains multiple samples for each individual. The matching process uses similarity metrics such as cosine similarity or Euclidean distance to determine how closely the detected face corresponds to known identities. If the similarity score exceeds a predefined threshold, the system successfully identifies the individual.

Upon successful recognition, the system automatically records the attendance of the identified individual. This information is stored in a structured CSV file, which includes details such as the individual's name or ID, along with the date and time of detection. This automated logging mechanism eliminates the need for manual intervention and ensures accurate and reliable attendance records. An important feature of the proposed system is its ability to operate independently without requiring continuous internet connectivity. All computations, including face detection, feature extraction, and data storage, are performed locally on the device. This not only enhances system efficiency by reducing latency but also ensures data privacy, as sensitive biometric information is not transmitted to external servers.

Furthermore, this system is designed with a integrated button at the sunglasses. A manual activation switch is integrated into the sunglasses, allowing the user to start and stop the system as needed. This feature provides flexibility and unnecessary and unwanted data capture and computational use. This is made to align with ethical considerations related to user consent and privacy.

In addition to its functional capabilities, the system is optimized for real-time performance. By using a lightweight version of the YOLOv9 model and efficient processing techniques, the system achieves high frame rates, enabling smooth and uninterrupted operation. This ensures that multiple individuals can be detected and recognized quickly, making the system suitable for environments such as classrooms, offices, and public gatherings.

Overall, the proposed system represents a significant advancement over traditional attendance methods by combining portability, automation, and intelligence into a single wearable device. Its modular architecture allows for easy upgrades and scalability, making it adaptable to future improvements in both hardware and software technologies. This system not only enhances operational efficiency but also sets the foundation for the integration of advanced AI solutions into everyday wearable devices.

#### **4.1.1 Purpose and Design Philosophy**

The proposed solution is a self-contained, wearable framework that merges hardware and software for real-time attendance tracking. Unlike static systems, this design offers a user-friendly, mobile alternative by embedding all logic within smart sunglasses. The goal is to provide automated tracking that integrates seamlessly into natural environments. In many educational institutions and companies, attendance is still reported manually or via semi-automated technologies, which require additional user actions. Moreover, these technologies can be time-consuming when used in busy environments with a large number of participants. In some instances, these methods might even be prone to manipulations and inaccuracies, including proxy attendance and incorrect data registration. As can be seen from the example above, there is a need to improve current attendance technologies by incorporating advanced solutions in a more convenient form.

The design philosophy of the proposed project is based on three core principles, namely portability,

automation, and privacy-awareness. These qualities will be ensured by embedding the system into the wearable accessories, automating attendance handling with the help of face detection and recognition algorithms, and conducting local operations to prevent unnecessary communication with internet-based infrastructures.



*Fig. Spectacles integrated with micro camera and a button*

#### **4.1.2 General Working Principle of the System**

The basic working principle of the system is connected to the constant acquisition of visual data and intelligent analysis. Once the device is activated, the embedded camera will start acquiring video frames from the surrounding environment. Afterward, the video streams will be uploaded to the processing unit to initiate further operations.

The first stage will include face detection. During this procedure, the system will scan the video frames and detect whether one or more faces are available. The next step will include face localization and isolation, meaning that the region of interest (ROI) will be located within a bounding box and removed from the rest of the image to enable further processing.

Afterward, the facial feature extraction will be conducted to convert each face into a set of values. To be precise, the system will transform each face from an image format to a set of machine-readable features, which will serve as unique identity signatures. Then, the extracted features will



be compared to the previously stored representations to determine whether the detected face belongs to a particular participant.

The system captures data through a nose-bridge-mounted micro HD camera, ensuring a frontal perspective that aids detection accuracy. All logic is executed by a Raspberry Pi, which balances cost, energy use, and processing power.

Once the similarity score is high enough to confirm the match, the identity will be recognized and attended. The attendance data will be logged into the CSV file with information regarding the date and time of detection. Thus, the proposed system operates as a closed-loop pipeline, transforming visual input data into recognizable identities and storing attendance records automatically.

#### **4.1.3 Major Functional Components of the System**

To obtain a complete understanding of the proposed system, it is vital to identify its functional components. As mentioned above, the proposed system will consist of several interrelated modules that are responsible for the operation of distinct processes. The first module will be the image acquisition module, which will be responsible for capturing live visual data using the micro HD camera. This component will influence the quality of the raw input and, thus, affect the performance of the following procedures.

The second module will be the face detection module, which will be based on the YOLOv9 architecture. This stage will ensure that only faces are processed by the recognition algorithm.

The third module will be the feature extraction module, which will be responsible for converting each detected face into a machine-readable format. At this stage, the face will be transformed into a set of unique features, which will be used to identify the participant's identity.

The fourth module will be the matching and recognition module, where the extracted facial features will be compared with the stored database of registered users. Based on the obtained similarity scores, the system will decide whether the participant belongs to a previously enrolled identity.

Finally, the last module will be the attendance logging module, where the recognized individual's data will be recorded into the CSV file in an organized way.

Overall, these components will create a well-organized pipeline, where the effectiveness of the recognition algorithm will be achieved through the synergy between the modules.

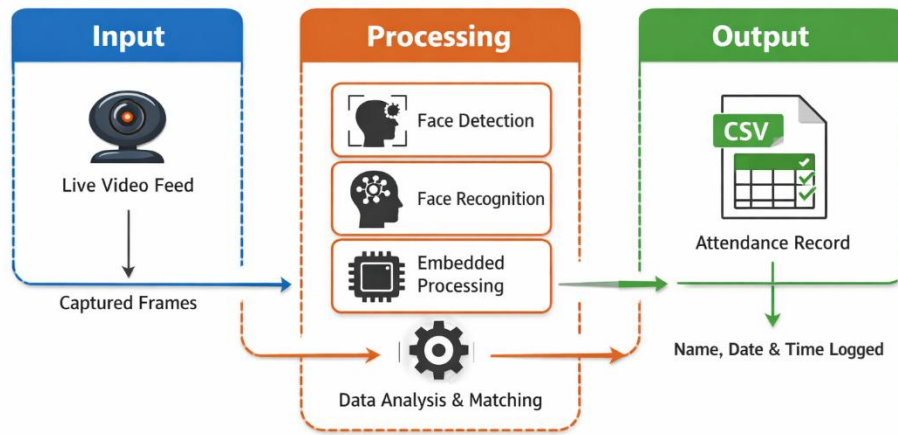
#### **4.1.4 Workflow of the Proposed System**

It is also crucial to outline the workflow of the proposed system. Specifically, it will be a sequential process consisting of multiple stages. First, the wearable device will be turned on by pressing the activation button. Second, the micro HD camera will start capturing live videos from the surrounding environment. Third, the acquired frames will be uploaded to the embedded processing unit. Fourth, the YOLOv9 algorithm will be responsible for detecting the presence of faces in the frames. Fifth, the ROI will be cropped and forwarded to the face recognition module. Sixth, each face will be transformed into a feature vector. Seventh, the feature vector will be compared with the stored database of registered users. Eighth, if the similarity score is higher than a pre-defined value, the identity will be matched and recognized. Ninth, the identified individual's data will be logged into the CSV file with detailed information about the name or identity label, date, and detection time. Finally, the recognition procedure will continue until the device is deactivated.

#### **4.1.5 Input, Processing, and Output Structure**

Alternatively, the structure of the proposed system can be viewed in terms of input, processing, and output. The input of the proposed system will be live video frames captured by the camera. This will be the primary raw input necessary for further processing. The input quality will be influenced by the position of the camera.

The processing of visual data will include all operations aimed at the transformation of the input data into recognizable faces and identification of identities. This process will be managed by the embedded processor and performed in a resource-limited environment. Finally, the output will be an attendance record created by the system once the recognition was successful. The results will be stored in a CSV file and include the name of the participant, date, and detection time.



*Fig. Architecture of the model*

#### 4.1.6 Real-Time Nature of the Proposed System

An essential feature of the proposed system is its real-time operational capabilities. When discussing attendance recognition, speed becomes a critical factor because delayed detection will make the system inconvenient and impractical for deployment. Thus, the attendance technology cannot be slow if one wants to apply it in crowded places like classes and offices.

The presented system will be able to process incoming video frames in real time due to the use of the YOLOv9-S model. In addition to the satisfactory accuracy level, the model will guarantee the fastest speed and will be light enough to operate in the resource-limited environment. Hence, the system will be capable of processing incoming visual data promptly and detecting faces almost immediately.

Moreover, the real-time capability of the proposed model will provide another advantage. Specifically, it will ensure that the attendance process can be conducted in a natural way without forcing participants to stay still. Thus, the proposed system can be used in mobile scenarios and support the convenience of attendance reporting.

#### 4.1.7 Portability and Wearable Integration

The primary distinguishing feature of the proposed project is the utilization of wearable integration.

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Sunglasses with YOLO V9 Technology

Most current attendance systems rely on stationary designs and, thus, lack flexibility. Instead, the presented model will integrate the attendance mechanism into portable sunglasses equipped with a micro HD camera and a dedicated processing unit.

Several benefits will follow from this design decision. First, the system will be mobile, facilitating further use cases. Second, the participant will not have to interact with the device explicitly. Instead, the system will capture his/her face and conduct further operations automatically. Lastly, the sunglasses' design will provide an optimal angle for the detection procedure because the camera will be facing forward.

However, the incorporation of the proposed model into the wearable accessories will pose several technical challenges. In particular, this design will require optimizing the hardware to reduce the size of the processor and batteries while maintaining sufficient performance. Also, the system will have to address the issues related to user comfort. Nevertheless, the presented design will ensure that the project will remain consistent with its objectives.

#### **4.1.8 Offline Operation and Data Privacy**

Another distinctive feature of the proposed system will be the possibility of offline operation. All major functions, including face detection, recognition, and attendance logging, will be conducted locally on the device. This feature will increase efficiency since data transfer to the external server will no longer be necessary.

At the same time, offline processing will contribute to data privacy protection. The use of biometric information for identity detection poses significant security risks because sensitive data is constantly transmitted via network infrastructures. In turn, the proposed system will keep the entire process local and eliminate any potential threats to user privacy.

Therefore, the offline operation will become the main advantage of the proposed model, making it safe, efficient, and practical.

#### **4.1.9 Significance of the Overall System Design**

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The significance of the proposed system design is that it integrates multiple emerging trends. In particular, the framework will incorporate wearable technology, embedded hardware, real-time face detection, and attendance recognition into one solution. Thus, the project will demonstrate a new way of combining multiple concepts to create a powerful technological tool.

On the one hand, the system will represent a state-of-the-art technology that combines advanced models with a compact design. On the other hand, it will provide a useful application of machine learning models in everyday activities. Thus, the project will prove that emerging innovations can be successfully implemented in practice.

## 4.2 Hardware Design and Implementation

The hardware design of the proposed system plays a crucial role in enabling real-time facial recognition within a compact, wearable form factor. Unlike conventional systems that rely on bulky setups or fixed installations, this system is engineered to be lightweight, portable, and seamlessly integrated into a pair of sunglasses. The primary objective of the hardware design is to ensure efficient processing, reliable performance, and user comfort while maintaining the compactness required for a wearable device.

The system is built by integrating multiple hardware components, each carefully selected based on performance, size, power consumption, and compatibility with embedded systems. These components work together to create a self-contained unit capable of capturing, processing, and storing data in real time.

At the front end of the system is the **micro HD camera**, which serves as the primary input device. This camera is capable of capturing video at a resolution of 720p, which provides a balance between image quality and processing efficiency. Higher resolutions, while offering more detail, would significantly increase computational load and reduce system performance. The camera is strategically mounted at the nose bridge of the sunglasses, ensuring that it captures a natural, forward-facing view. This positioning is critical, as it maximizes the probability of capturing clear and frontal facial images, thereby improving the accuracy of the detection and recognition

processes.

The captured video data is processed by the **Raspberry Pi**, which functions as the central processing unit of the system. The Raspberry Pi is selected due to its compact size, affordability, and sufficient computational capability to run lightweight deep learning models. It is capable of handling tasks such as video frame extraction, face detection using the YOLOv9 model, and execution of the recognition pipeline. Although it does not match the performance of high-end GPUs, its efficiency and low power consumption make it an ideal choice for embedded and portable applications. Additionally, the availability of extensive software support and compatibility with Python-based frameworks further simplifies system development and deployment.

To support offline functionality, the system incorporates **local storage**, which is used to store facial datasets, feature embeddings, and attendance records. This storage is typically implemented using a microSD card or onboard flash memory. By storing data locally, the system eliminates the need for continuous internet connectivity, which not only improves performance by reducing latency but also enhances data privacy. Sensitive biometric data remains within the device, minimizing the risk of unauthorized access or data breaches.

Power management is another critical aspect of the hardware design. The system is powered by a lithium-polymer (Li-Po) battery, which is chosen for its high energy density and lightweight properties. The battery provides sufficient power to operate the camera, processing unit, and other components for extended periods. Efficient power utilization is essential to ensure that the system remains functional throughout its intended usage duration. In future iterations, power optimization techniques such as dynamic frequency scaling and sleep modes could be implemented to further extend battery life.

To provide user control and improve usability, an activation switch is integrated into the sunglasses frame. This push-button mechanism allows the user to manually start or stop the system as needed. This feature is particularly important from both a usability and ethical perspective, as it prevents continuous or unintended data capture. Users can activate the system only when attendance needs to be recorded, ensuring better control over data collection.

A significant aspect of the hardware implementation is the physical integration of all components into the sunglasses frame. This is achieved using 3D printing technology, which allows for customized design and precise placement of components. The frame is designed with dedicated compartments and slots to securely hold each hardware element in place. Special attention is given to weight distribution to ensure that the device remains comfortable for extended use. Improper weight balancing could lead to discomfort, making the system impractical for real-world applications.

The design also considers factors such as heat dissipation and wiring management. Since the Raspberry Pi and battery can generate heat during operation, adequate ventilation or heat-resistant materials are considered in the frame design. Similarly, internal wiring is arranged in a compact and organized manner to prevent interference with the user's comfort or the system's functionality.

Another important consideration in the hardware design is durability and robustness. The system is intended for regular use in environments such as classrooms or workplaces, where it may be exposed to movement and minor impacts. Therefore, the frame and component mounting are designed to withstand such conditions without compromising performance.

Overall, the hardware design represents a careful balance between functionality, performance, and usability. By integrating all necessary components into a compact and wearable form factor, the system achieves a high level of portability while maintaining the computational capability required for real-time facial recognition. This design not only supports the technical requirements of the system but also ensures that it can be practically deployed in real-world scenarios.

The successful implementation of this hardware architecture forms the foundation for the software components and recognition pipeline, enabling the system to operate efficiently as a complete, standalone solution.

### **4.3 Facial Dataset Preparation**

As stated above, the first and foremost step in creating an efficient facial recognition system is the

acquisition of a high-quality dataset of facial images for each participant. The accuracy, robustness, and generalization capabilities of the proposed model will depend on the quality of the provided data. Therefore, it is vital to ensure that the data acquisition process adheres to all the requirements to make it possible for the model to recognize participants correctly.

The data preparation process involves collecting facial images of participants. For that purpose, it is essential to consider the use of devices available for most people, such as smartphones. It should be noted that images will be collected at high enough resolution to ensure accurate recognition while keeping the size of images relatively small to enable efficient calculations.

One of the main objectives that should be accomplished during data collection is to increase the variety of the database. It is common knowledge that human faces can change depending on several factors, such as head position, expression, environmental lighting, and occlusion, among others. Consequently, the images collected will differ from each other regarding several characteristics, such as pose, expression, lighting conditions, and background.

For example, it is assumed that the poses of individuals in the dataset will differ significantly because there is no possibility of forcing individuals to present themselves only in the frontal position. Thus, images captured in slightly rotated or tilted positions will also be included in the dataset. Similarly, to avoid bias associated with a particular facial expression, images reflecting different expressions will be captured.

Additionally, variations in lighting conditions will be included in the dataset. For instance, images will be collected in bright sunlight, artificial lighting in offices, and low-light conditions to avoid making the model prone to the specific lighting conditions. Another type of variation that should be accounted for is occlusion, such as sunglasses. The proposed system also uses sunglasses; therefore, it is important to prepare a dataset that will teach the model how to recognize individuals even when wearing them.

After images are collected, they are classified into several folders corresponding to each person's unique identifier. Each folder will contain images belonging to one individual. As a result, each



folder will contain images corresponding to different angles, expressions, and backgrounds. The data preparation step is critical because it ensures that there will be no errors when labeling the images during the training phase.

Next, some preprocessing procedures will be applied to the images. For example, images will be resized to ensure the required resolution and normalized to eliminate noise. In addition, images will be transformed into grayscale format. Such data normalization is crucial because it ensures that all input images correspond to one format, which will facilitate efficient recognition during the training stage.

Data augmentation techniques will be applied to increase the size and variety of images. Several transformations, such as rotation, cropping, flipping, and changing brightness, will be used to enrich the database with additional information. For example, slight image rotation will be helpful in training the model to identify individuals who tilt their heads, and changing the level of brightness will account for different lighting conditions.

Class balancing is also critical during data preparation. In the proposed system, it is assumed that each class will contain approximately the same number of images. However, it might be challenging to capture the same number of images for each individual in reality. For that reason, data balancing should be performed to ensure that the model does not develop a bias toward the dominant class.

Feature embedding generation is an inseparable part of data preparation. During the initialization stage, all images will be processed using the feature extraction model, which will extract features and generate their numeric representations. In turn, these embeddings will be saved and used during subsequent stages to compare faces.

During the practical implementation of the proposed system, the data preparation process will be scaled and simplified to enable the convenient addition of new participants to the model. Therefore, it will be sufficient to collect images for each participant and store them in the designated directories. There will be no need to perform tedious data training procedures repeatedly.

At the same time, efforts will be made to avoid data redundancy, and images will be selected to maximize the variety in the dataset rather than increasing the volume of the data. Ethical considerations will also be taken into account when preparing the dataset. Specifically, data privacy and confidentiality requirements will be met when acquiring, storing, and analyzing images.

#### **4.4 YOLOv9 Model Architecture**

The efficiency and accuracy of the proposed facial recognition system depend greatly on the performance of the face detection model implemented in the initial stage of the process. For that reason, YOLOv9-S (You Only Look Once version 9 – Small) will be utilized as the face detection method. YOLOv9 can be considered the newest improvement of one of the best-performing object detectors available. It was created specifically to provide an optimal combination of speed, accuracy, and computational efficiency, which makes it a perfect choice for use in real-time applications on embedded hardware platforms like Raspberry Pi.

Unlike many conventional approaches that rely on multi-stage pipelines (first, potential detection regions are determined, and then their class labels are predicted), YOLOv9 uses a single-step detection strategy that enables simultaneous localization and classification. Such an architecture not only improves inference time but also allows detecting the objects instantly.

From a technical perspective, YOLOv9 can be referred to as a regression algorithm. That means that the model detects objects not as multiple stages but via predicting the locations of their bounding boxes and probabilities associated with each of the detected classes. This characteristic gives the model an opportunity to consider the entire picture as one, making the process of recognizing objects more accurate.

The architecture of YOLOv9 can be subdivided into three main components. They have been developed for solving specific tasks within the detection pipeline. The backbone part of the model serves the function of extracting features from the image. As in all other models of this family, the backbone network used for implementing this algorithm is CSP Darknet-based. This architecture allows improving the flow of gradient information and reducing the number of repeated calculations

that occur during training.

When processing the input, early layers are responsible for extracting low-level features that correspond to simple edges, colors, and textures of the image. Deeper layers extract high-level representations, like face features, patterns, and shapes. As a result, the model can identify various types of objects.

CSP stands for Cross Stage Partial network and means that the feature maps split into two streams in the architecture and later merged to increase learning effectiveness. Thus, YOLOv9 benefits from the reduced model size and increased performance. Another part of the model's architecture is the neck module. Its function is to aggregate the extracted features and facilitate the detection of objects. For this purpose, YOLOv9 uses PANet – a Path Aggregation Network.

The primary goal of the neck is to combine features extracted from several layers, which helps the algorithm detect objects on different scales. Indeed, the distance between the camera and the face varies, causing faces to appear either smaller or larger than usual. As a consequence, the face detection algorithm must be able to recognize faces on various sizes. Multi-scale detection is provided by the use of a combination of spatial and semantic features obtained by the model.

The last stage of the YOLOv9 architecture is called the detection head. The task of this part of the model is generating detections, which include the bounding box coordinates, class labels, and the probability value assigned to each object.

As mentioned above, the model is designed for recognizing faces, so only one type of object can be detected in this case. YOLOv9 has been improved compared to previous versions of YOLO in terms of the detection head and its architecture. Most importantly, the algorithm uses an anchor-free detection approach, which means that the algorithm predicts the location of the detected object directly rather than relying on predefined anchor points. Also, improved bounding box regression enables detecting objects regardless of their positioning.

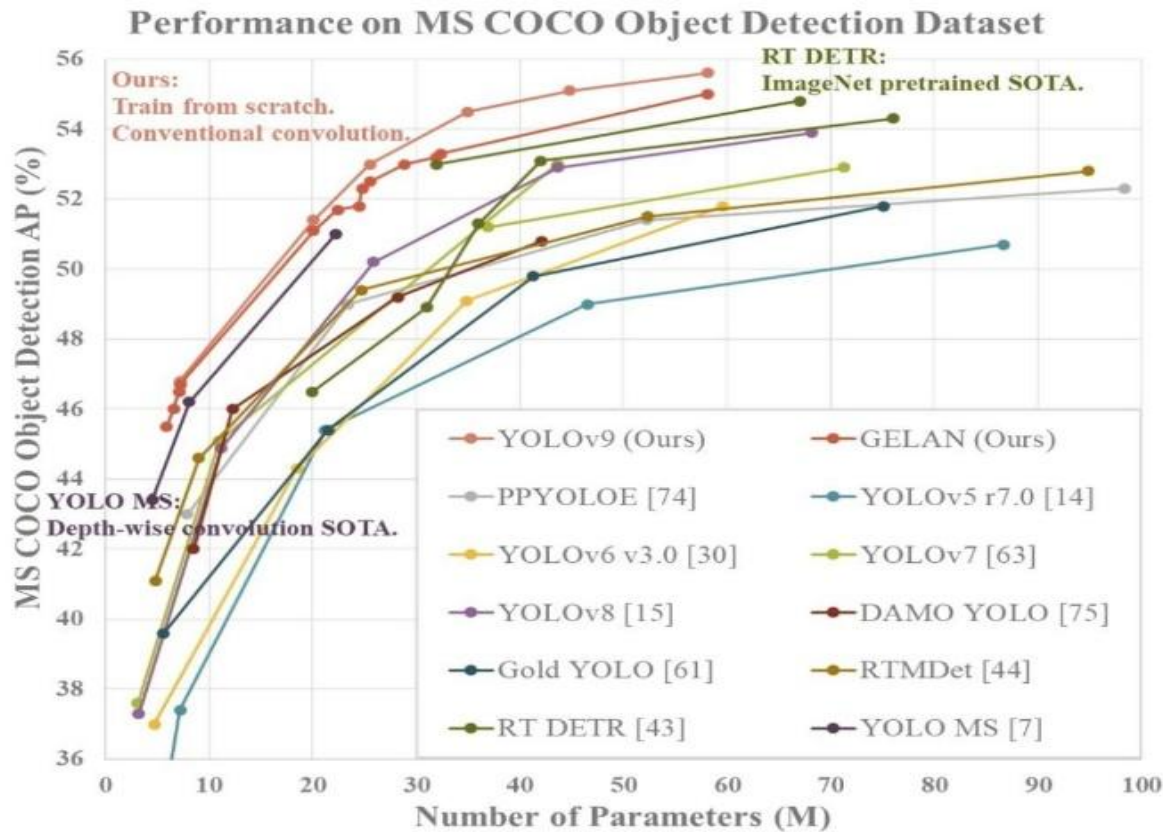


Fig 2. YOLOv9 comparison graph

As can be seen from the information provided above, YOLOv9 is a model that performs quite well in real-time applications due to high speed and accuracy. YOLOv9-S is one of the versions of this model optimized for lightweight deployment. Other variants of the model (YOLOv9-M and YOLOv9-L) provide enhanced accuracy; however, their deployment is less effective due to higher computational costs, which is not desirable for implementation on the Raspberry Pi. Thus, YOLOv9-S can be used for achieving appropriate inference speed without sacrificing accuracy.

The process of using the detection model is rather straightforward and does not contain complex procedures. The first step is preprocessing of the image – adjusting its dimensions to match the input requirements and performing normalization. Then, the image goes through the neural network, whose output consists of multiple detection proposals. For each of these predictions, a certain confidence value is given that shows how likely it is a real object.

Thresholding and Non-Maximum Suppression techniques are utilized to obtain only the necessary output data. Thresholding eliminates predictions that do not meet the specified requirements

(usually, the lowest allowed confidence value). The remaining proposals can overlap one another and show multiple faces in a single bounding box, which leads to applying the suppression technique. Non-maximal suppression is performed until there are no overlapping bounding boxes. One of the greatest advantages of using YOLOv9 is the robustness of the model. It is capable of identifying objects in adverse conditions – in poor lighting or when the subject does not face the camera and wears glasses. Thanks to deep feature extraction, it is possible to find partial faces and determine whether they are indeed humans.

From a computational standpoint, the model can be optimized for running on the Raspberry Pi by implementing various optimization techniques, such as compression and memory management. Despite the limited computing power of this platform, YOLOv9 can operate with the rate of up to 48–50 FPS, which is quite satisfactory.

One more benefit of using this algorithm is related to its modularity. In future, the algorithm can be modified easily due to its architecture, and its detection abilities can be improved significantly. The only stage that should be changed is the face recognition model itself because other steps are universal and can remain unchanged.

From a practical standpoint, integrating the detection model into the recognition pipeline does not create any difficulties for a developer. The frames from the camera go through the neural network sequentially, and their output is forwarded to the next stage right after processing.

#### **4.5 Face Recognition Pipeline**

Face recognition is an essential component of the proposed wearable attendance system. In essence, the pipeline is responsible for the transformation of raw data inputted into the identification of an individual based on the detected face. Furthermore, the pipeline generates a log record about the recognized individual that contains their name, identification number, date, and time of detection. Thus, the success of the entire proposed system will rely on the efficiency of this pipeline.

The face recognition pipeline comprises several stages of face detection, face localization and cropping, preprocessing, feature extraction, feature matching, identity recognition, and attendance logging. Each of the stages described above is responsible for completing a particular task. In this regard, each of the stages forms a pipeline that transforms raw input into valuable output

information.

The first stage of the pipeline consists of image acquisition. In this case, the system captures live video frames through a micro HD camera integrated into the sunglasses. The camera streams each of the captured frames to the processing unit continuously in order to provide real-time data input. Therefore, the quality and position of the camera are crucial at this stage because they affect the quality of the subsequent process.

After acquiring the live frame, the face recognition pipeline continues with the face detection stage. At this point, the entire frame is inputted into the YOLOv9 model. After that, the YOLOv9 model identifies regions of interest in the captured frame containing human faces. The identified regions are denoted by bounding boxes with a certain level of confidence associated with them. All low-confidence detections are filtered out, meaning that only confident detections are forwarded to the following stages.

Once the faces are detected, they go through the face localization and cropping stage. In this regard, each of the detected faces is localized and extracted from the original frame using the provided bounding boxes. As a result, the faces are isolated from other visual elements such as backgrounds and unrelated objects. This process is essential since it filters out unnecessary information from the image and makes the further stages of the pipeline more efficient.

The cropped face image is passed to the next stage of the pipeline. Specifically, the cropped image undergoes preprocessing, thus preparing the image to be fed into the feature extraction model. For example, the preprocessing stage includes such actions as cropping the face region from the original image, adjusting the image size, and normalizing the image pixels. These procedures make the pipeline more efficient in the further process.

Once the image is preprocessed, it moves to the feature extraction stage. At this point, the model takes each preprocessed face image and transforms it into a numerical feature vector or embedding. This stage of the pipeline is highly important since it determines the accuracy of the whole proposed system. Feature extraction models, such as the Siamese network or embeddings model, are utilized

at this stage. Embeddings are multidimensional vectors that represent unique characteristics of an individual's face. Compared to raw images, embeddings have numerous advantages. First, they allow performing comparisons in real-time due to their compactness. Second, embeddings are designed in such a way that the distance between embeddings of the same person is small, while the distance between embeddings of two different people is large.

After obtaining the feature vector or embedding, the system continues with the feature matching stage. At this point, the extracted feature vector is compared against a database of pre-stored embeddings. These embeddings were obtained previously in the dataset preparation stage and are used as references when identifying an individual.

There are several similarity metrics available for comparing feature vectors, including cosine similarity and Euclidean distance. The former measure estimates the angular distance between two vectors, while the latter measures the direct distance between them in the feature space. A similarity score is estimated based on the similarity metric used. This score indicates the degree of similarity between the two vectors.

If the similarity score between the detected face and one of the database embeddings exceeds a certain threshold, the individual is identified, and their name is extracted. Otherwise, the system recognizes the individual's face as an unknown entity. The similarity threshold is highly crucial since it influences the accuracy of the pipeline and the occurrence of false recognition.

Having successfully recognized the individual, the system obtains their label and forwards it to the next stage of the pipeline. In this context, the label can contain either the name or identification number of the individual. The last stage of the face recognition pipeline consists of attendance logging. This stage involves generating a CSV file containing the information related to the detected individual.

In particular, the CSV file records the name or identification number of the detected person, as well as the date and time of detection. The process is completely automated, thus eliminating the necessity of human involvement. Besides, there are measures in place that prevent the duplication of the same individual's detection, such as checking the attendance status of the individual in the

current period.

As previously mentioned, the proposed system possesses some inherent characteristics that should be discussed. First, the face recognition pipeline is capable of processing multiple faces simultaneously. In this respect, if multiple individuals are detected, they are all processed independently in order to recognize each individual accurately.

Second, the face recognition pipeline possesses a high level of reliability. In this context, various measures are put in place to eliminate errors and increase the accuracy of the process. Such measures include the use of confidence thresholds, similarity thresholds, and data filtering.

Third, the face recognition pipeline can be deployed efficiently due to the optimization of code using appropriate libraries and frameworks. Furthermore, the face recognition pipeline uses efficient memory allocation and processing techniques that allow executing the pipeline on Raspberry Pi devices. Fourth, the proposed system is secure and protects users' personal information. In this regard, all processing occurs locally on the device, preventing the transmission of face data to external systems.

## 4.6 Implementation Details

This section presents the core implementation of the proposed facial recognition-based attendance system. The code is developed using Python and integrates multiple libraries such as OpenCV, face\_recognition, NumPy, and the Ultralytics YOLO framework.

The implementation follows the pipeline described in the previous sections, including face detection, feature extraction, identity matching, and attendance logging. The following code snippets highlight the key components of the system.

### Algorithm

#### 1. Face Detection and Recognition

Input: Live video stream from camera

Output: Identified face with corresponding name or "Unknown"



1. Start system
2. Initialize camera and load YOLO model
3. Load known face encodings and labels
4. Capture video frame from camera
5. Convert frame to RGB format
6. Apply YOLO model to detect faces
7. For each detected face:
  - a. Extract bounding box coordinates
  - b. Crop face region from frame
  - c. Apply preprocessing (resize, normalize)
  - d. Generate facial encoding using feature extraction model
  - e. Compare encoding with stored face encodings
  - f. Compute similarity score using distance metrics
  - g. If similarity score exceeds threshold:
    - Assign corresponding name
    - Else:
      - Assign label as "Unknown"
8. Display bounding box and name on frame
9. Repeat steps 4–8 until system is stopped
10. End system

## 2. Attendance Logging

Input: Recognized face name

Output: Entry stored in attendance file

1. Receive recognized name from recognition module
2. Open attendance CSV file in read/write mode
3. Read existing entries from file
4. Check if name already exists in current session
5. If name does not exist:
  - a. Get current date and time
  - b. Append name, date, and time to file
6. Else:
  - Skip entry to avoid duplication
7. Close file
8. Return confirmation

## Snippet

```
import cv2
import face_recognition
import numpy as np
from ultralytics import YOLO
from datetime import datetime
```

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```
# YOLO Model
yolo = YOLO("yolov8n-face.pt")

# Load Face Database
known_face_encodings = []
known_face_names = []

def load_known_faces():
    import os

    base_path = "faces/"

    for person_name in os.listdir(base_path):
        person_path = os.path.join(base_path, person_name)

        if not os.path.isdir(person_path):
            continue

        for img_name in os.listdir(person_path):
            img_path = os.path.join(person_path, img_name)

            img = face_recognition.load_image_file(img_path)
            encodings = face_recognition.face_encodings(img)

            if len(encodings) > 0:
                known_face_encodings.append(encodings[0])
                known_face_names.append(person_name)

load_known_faces()

# Attendance Marking
def mark_attendance(name):
    with open("Attendance.csv", "r+") as f:
        lines = f.readlines()
        names_in_file = [line.split(",")[0] for line in lines]

        if name not in names_in_file:
            now = datetime.now()
            time = now.strftime("%H:%M:%S")
            date = now.strftime("%d-%m-%Y")
            f.write(f"{name},{date},{time}\n")

# Webcam / Sunglasses Feed
cap = cv2.VideoCapture(1)
```

```

print("System Started... Press 'q' to Quit")

while True:
    ret, frame = cap.read()
    rgb_frame = cv2.cvtColor(frame, cv2.COLOR_BGR2RGB)
    if not ret:
        break

    # YOLO detections
    results = yolo(frame, verbose=False)

    # Process detections
    for r in results:
        for box in r.bboxes.xyxy.cpu().numpy():
            x1, y1, x2, y2 = map(int, box)
            # Clamp to image bounds
            h, w = frame.shape[:2]
            x1 = max(0, min(x1, w-1))
            x2 = max(0, min(x2, w-1))
            y1 = max(0, min(y1, h-1))
            y2 = max(0, min(y2, h-1))

            # Optionally expand the box a bit (padding) for better landmarks
            pad = 10
            x1p = max(0, x1 - pad)
            y1p = max(0, y1 - pad)
            x2p = min(w-1, x2 + pad)
            y2p = min(h-1, y2 + pad)

            # Use the original RGB frame + location in (top, right, bottom, left)
            top, right, bottom, left = y1p, x2p, y2p, x1p
            face_locations = [(top, right, bottom, left)]

            # Get encodings for that location – returns list (may be empty)
            encodings = face_recognition.face_encodings(rgb_frame, face_locations)

            if len(encodings) == 0:
                # No descriptor found – skip
                name = "Unknown"
            else:
                face_enc = encodings[0]
                # Compare with known faces
                matches = face_recognition.compare_faces(known_face_encodings,
face_enc, tolerance=0.5)
                face_distances = face_recognition.face_distance(known_face_encodings,
face_enc)
                if len(face_distances) > 0:
                    best_match = np.argmin(face_distances)
                    if matches[best_match]:
                        name = known_face_names[best_match]

```

```
        mark_attendance(name)
    else:
        name = "Unknown"
    else:
        name = "Unknown"

    # Draw box + name on the original BGR frame
    cv2.rectangle(frame, (x1p, y1p), (x2p, y2p), (0, 255, 0), 2)
    cv2.putText(frame, name, (x1p, y1p - 10),
                cv2.FONT_HERSHEY_SIMPLEX, 0.7, (0, 255, 0), 2)

cv2.imshow("YOLO Face Recognition Attendance", frame)

if cv2.waitKey(1) & 0xFF == ord("q"):
    break

cap.release()
cv2.destroyAllWindows()
```

## **CHAPTER 4**

### **RESULTS**

## **RESULTS**

This section evaluates the wearable system based on accuracy, real-time speed, and environmental resilience. Tests were performed using the hardware prototype and a customized dataset. The results are evaluated based on a variety of metrics, such as detection and recognition accuracy, performance in real-time conditions, robustness of the system, and overall speed of the proposed solution. The results presented in this section are based on the experiments performed with the help of the developed hardware prototype and personalized dataset.

In order to test the performance of the proposed system, tests were conducted in both controlled and real-world conditions. The dataset included images of individuals shot from various angles, in different poses, with different lighting, and with different facial expressions. The implementation of the system was conducted on the Raspberry Pi microcontroller, and its performance was tested in real-time situations with natural movement of individuals in the camera's field of view.

Recognition accuracy is one of the key performance parameters of the developed system. With respect to its recognition accuracy, the system has managed to achieve approximately 90% recognition rate. Such recognition rate indicates high accuracy in recognizing faces of people. In this case, a very important role is played by the fact that the database includes multiple images per individual and thus allows for obtaining an accurate representation of their faces.

In addition, upon a more detailed analysis of recognition rate performance, one can determine that it is positively correlated with such aspects as lighting and frontal position of the head. Under the described circumstances, the system performs with almost perfect recognition. In some cases, with partial rotation or inclination of the head, however, the performance decreases slightly. Nevertheless, the presence of such poses in the dataset makes the system robust enough for proper functioning.

Apart from that, another metric worth considering is processing speed measured in FPS. As it has been mentioned previously, the implemented YOLOv9-S model performs with approximately 48–50 FPS rate, which is quite good for a real-time application. As such, it is possible to claim that the

system performs in real time. Another aspect considered as part of the performance analysis is face detection. The implemented YOLOv9 model shows great efficiency in face detection. For instance, it can recognize faces at different ranges and scales within a single image.

Moreover, the ability of the model to work in multiple scales allows detecting faces at both near and relatively far distances. In addition, it is worth pointing out that the system can detect and process multiple faces at once. This feature is especially useful in such real-world settings as classrooms or conferences where multiple users are to be recognized.

An important aspect tested for the developed system is its performance under occlusion condition, which appears due to wearing sunglasses in conjunction with the system. As expected, the model performs quite well with partially visible faces because the deep learning approach allows recognizing people by their partially visible face features. Lighting performance is another important criterion, and according to the results, the system performs well under indoor and outdoor lighting. When it comes to extreme low lighting, however, a slight decrease in detection and recognition rate takes place. Such a performance is to be expected, though, because low lighting conditions reduce the quality of face images being recognized by the model.

Another important criterion is the efficiency of attendance logging process. As it has been shown in the test results, the attendance data can be automatically logged into the CSV file without any additional user interaction required. Thus, the system records name of the person, his/her ID number, as well as the exact day and time.

The last aspect taken into consideration is the latency of the process since the time interval between face detection and face recognition is rather important for the overall performance of the system. Thus, it was noted that there is virtually zero latency in the performance of the system, and the system responds immediately after face detection.

With respect to usability of the developed system, it is evident that its design is rather successful because of no need for standing still in front of a static camera and using the activation switch. In other words, the user can control the attendance process easily and without compromising privacy. However, despite numerous positive performance metrics, several limitations were also noted. The

system shows worse accuracy performance under extreme lighting, movement, and blurriness. In addition, performance of the system depends heavily on the dataset available since the more varied images, the better accuracy.

Another aspect considered in terms of limitations is the computational power of the hardware device. It should be noted that the system works successfully on the Raspberry Pi device, but the increase of complexity of the algorithm may affect performance. Overall, the proposed system shows great performance in terms of real-time face detection and recognition. It can detect multiple faces and faces in varying conditions with an adequate speed rate and acceptable accuracy level. Thus, the results show that the system is practically effective and ready to be utilized.



## **CHAPTER 5**

### **DISCUSSION**

## **DISCUSSION**

The results obtained from implementing the proposed system make it possible to thoroughly evaluate its capabilities, advantages, and limitations. This section provides a detailed interpretation of the outcomes and connects them with the system's design choices and methodology.

The experimental findings validate the effectiveness of the proposed design. Reaching 90% accuracy confirms that the fusion of YOLOv9-based detection with an embedding-based pipeline is a viable strategy for embedded applications. The achieved performance can be attributed to the integration of YOLOv9-based face detection and embedding-based recognition. In this combination, detection ensures accurate localization of faces, while the embedding model enables effective identification of individuals.

A closer look at the results shows that dataset preparation plays a crucial role in achieving this accuracy. By including multiple images per individual under different lighting conditions, poses, and facial expressions, the system learns generalized facial representations. This reduces overfitting and improves performance in real-world scenarios where variability is unavoidable.

Another key strength of the system is its processing speed, which reaches approximately 48–50 FPS. Achieving this level of performance on a Raspberry Pi highlights the efficiency of the chosen YOLOv9-S model. The lightweight nature of the model allows it to operate with limited computational resources while still maintaining acceptable accuracy. This demonstrates the importance of selecting models that balance speed and performance, especially in embedded systems.

The high frame rate also enhances the overall user experience by enabling real-time operation. Faces are detected and recognized almost instantly, ensuring smooth and uninterrupted functionality. The absence of noticeable latency indicates that the system pipeline is well-optimized and free from major bottlenecks.

Another important observation is the system's robustness in handling occlusion. Despite the

presence of sunglasses, which partially cover facial features, the system maintains reasonable recognition accuracy. This suggests that the feature extraction model is capable of identifying individuals based on visible facial characteristics, rather than relying on complete facial information.

The system also performs well in multi-face scenarios, where multiple individuals are present within a single frame. This capability is essential for practical applications such as classrooms and workplaces. The use of YOLOv9 enables efficient detection and processing of multiple faces without significantly affecting performance. However, several limitations were identified during evaluation. One of the main challenges is sensitivity to lighting conditions. While the system performs well under normal lighting, its accuracy decreases in low-light environments. This is mainly due to reduced visibility, which affects both detection and feature extraction. Future improvements could involve image enhancement techniques or specialized hardware such as infrared cameras.

Motion blur is another factor that impacts performance. Rapid movement can result in blurred images, making it difficult for the system to extract clear facial features. Addressing this issue may require improvements such as faster shutter speeds or motion stabilization techniques.

The system's dependence on dataset quality is also evident. High accuracy is achieved when the dataset is diverse and well-structured. However, limited or poorly varied datasets can reduce performance. This highlights the importance of continuously updating and maintaining high-quality training data. From a design perspective, the decision to use local processing instead of cloud-based computation offers significant advantages. It reduces latency and enhances data privacy by keeping sensitive biometric information within the device. This aligns with modern edge computing approaches, which prioritize efficiency and security.

The wearable nature of the system introduces both benefits and challenges. On one hand, it provides flexibility and allows attendance to be recorded naturally without requiring users to interact with fixed devices. On the other hand, it introduces challenges related to camera alignment and user movement. The results indicate that these challenges have been managed effectively, although further improvements in ergonomics could enhance usability.

Another important consideration is the trade-off between model complexity and hardware capability. While more complex models could improve accuracy, they would also require greater computational resources. The choice of YOLOv9-S reflects a balanced approach, ensuring real-time performance without significantly compromising accuracy. The modular design of the system is another strength. Since each component operates independently, the system can be easily upgraded or modified. For example, more advanced recognition models can be integrated in the future without affecting the detection stage. In terms of real-world impact, the system demonstrates the feasibility of integrating advanced AI technologies into wearable devices. It opens up possibilities for applications beyond attendance tracking, such as security systems and personalized user interactions. Additionally, the automation of attendance reduces manual effort and improves efficiency in administrative tasks.

## **CHAPTER 6**

## **CONCLUSION**

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In this research, we have designed and developed a novel facial recognition based attendance tracking system using a pair of sunglasses as an innovative approach. The main aim of the study was to create a portable, efficient, and real-time system to solve issues related to conventional methods of taking attendance.

The integration between hardware and software components is successfully achieved by implementing this system. Since the device contains a micro HD camera, Raspberry Pi, and local storage facility, it becomes completely portable and independent of other devices and networks. In terms of software part, face detection with the help of YOLOv9-S model and further process for facial recognition with deep learning-based feature extraction technique ensures that individuals will be recognized with high efficiency and accuracy.

As far as performance of the proposed system is concerned, it provides an average accuracy around 90% in the recognition process and real-time speed ranges from 48 fps to 50 fps. This shows that the system works very efficiently under different circumstances and even performs robustly under partial occlusions which means that it is reliable in practice.

One of the unique features of this research is to implement facial recognition based attendance tracking in a wearable manner as compared to the conventional method of using stationary cameras. With this method, flexibility and portability aspects of the system become prominent and the whole procedure of logging attendance automatically becomes more efficient and error-free.

There are some aspects in the process of building up a system in which care should be taken to achieve high accuracy with less computational cost. Dataset preparation, model choice, and proper tuning of parameters for optimization are some important factors for developing a reliable system with high accuracy and acceptable performance.

## **CHAPTER 7**

### **FUTURE WORK**

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Despite the strong performance and reliability demonstrated by the proposed system, there are several opportunities for further improvement and expansion in future work. These enhancements can focus on increasing accuracy, improving robustness, and extending the system's applicability across different real-world scenarios.

One important area for improvement is performance under low-light conditions. The system currently experiences a slight drop in accuracy when lighting is insufficient, as facial features become less visible. This limitation can be addressed by applying image enhancement techniques such as brightness adjustment, contrast improvement, and noise reduction. Additionally, incorporating hardware components like infrared cameras or depth sensors could significantly improve performance in such environments.

Another area that requires attention is robustness against motion and blur. In real-world situations, users may move frequently, which can lead to blurred frames and reduced recognition accuracy. This issue can be mitigated by using higher frame-rate cameras, motion stabilization techniques, or advanced filtering methods. Analyzing multiple consecutive frames instead of a single image can also improve reliability.

Improving the dataset is another key factor in enhancing system accuracy. Increasing the number of images per individual and ensuring diversity in pose, lighting, and background conditions will help the model generalize better. The use of larger datasets and transfer learning techniques can further strengthen feature extraction capabilities. Additionally, implementing continuous learning mechanisms would allow the system to adapt to changes in user appearance over time.

Future work may also explore the use of more advanced deep learning models. While the current system uses lightweight architectures suitable for embedded devices, more powerful models such as transformer-based or hybrid approaches could improve recognition performance. To maintain efficiency, optimization techniques like pruning and quantization can be applied.

From a hardware perspective, upgrading to more powerful embedded platforms such as NVIDIA



Jetson devices or dedicated AI accelerators could enhance performance. Improvements in battery life and power management would also make the wearable system more practical for extended use. Scalability is another direction for future development. The current system is suitable for small-scale use, but it can be extended to support larger environments such as universities or corporate settings. This may involve integrating centralized data management and combining edge and cloud computing approaches while maintaining data privacy.

Security can also be improved by incorporating anti-spoofing mechanisms. Techniques such as liveness detection, blink detection, depth sensing, or thermal imaging can help prevent unauthorized access using photos or videos. Furthermore, combining facial recognition with other biometric methods like voice recognition, iris scanning, or gait analysis could create a more secure multi-modal system.

The wearable design itself can be refined to improve user comfort and usability. Reducing device size and weight, improving ergonomics, and enhancing durability will contribute to a better user experience. Efficient heat management is also an important consideration for long-term usage. On the software side, the system can be extended with a user-friendly interface or mobile application. Features such as real-time monitoring, analytics, and reporting tools would transform the system into a complete attendance management solution.

Finally, the system has potential applications beyond attendance tracking. It can be adapted for use in security systems, access control, smart environments, and personalized user interactions. At the same time, future research should consider ethical and privacy aspects, ensuring secure data handling and compliance with regulations.

In conclusion, the proposed system provides a solid foundation for future advancements in wearable biometric technology. By addressing current limitations and exploring these improvements, the system can become more accurate, scalable, secure, and versatile for real-world deployment.

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